An Analysis of Binary ML Classification for Weather Prediction

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Abstract

Cloud cover has been observed and studied throughout a significant portion of history, especially since they are very discernable and noticeable compared to other meteorological variables, even though it can be challenging to measure. There are several applications and utilizations derived from clouds, such as having a sub-discipline in physics (i.e., cloud physics), and being an integral aspect of weather forecasting, in which it can have effects on the weather even when there is no precipitation associated with the clouds, namely being radiative heating/cooling. In this study, several machine learning techniques, being support vector machines (SVM), k-nearest neighbors (KNN), XGBoost, AdaBoost, Bagging, Random Forest, artificial neural networks (ANN), long short-term memory (LSTM), convolutional neural networks (CNN), and bidirectional LSTM (BiLSTM), as well as a few new methods, being extended LSTM (xLSTM), circular dilated CNN (CDIL), and CNN-BiLSTM, are performed on an atmospheric dataset that includes several meteorological phenomena (including cloud cover) for 18 European cities from 2000 to 2010, in which these models will be evaluated on a 60/40, 70/30, and 80/20 train-test subset that utilizes a classification threshold of 50% cloud cover for the city of Basel, Switzerland. Results from the evaluation metrics (train accuracy, validation accuracy, test accuracy, F1 score, kappa, recall, precision, and area under the curve (AUC)) indicate that more simple machine learning techniques generally performed better than the complex new methods, possibly due to the simplicity of the dataset and the binary nature of this study.

Introduction

Clouds are one of the most visible and earliest studied meteorological phenomena [1], with entire sub-disciplines dedicated to their independent chemistry, physics, dynamics, and roles within several other processes. Even when not associated with any precipitation, cloud cover alone can have several significant short-term and climatological impacts. The most relevant of these involve radiative absorption across short and long wavelengths [2], [3], consequently leading to effects on temperature [4]. Therefore, implementations of cloud microphysical schemes have thus far been made through numerical weather prediction and in some machine learning (ML) applications [5], [6]. With ML only having been in meteorological use over the past decade, there is yet to be a dedicated multi-day forecasting application of any such technique. This is what we seek to produce here; through the use of *[methodology undecided as of now, pending what we learn in class]*, we will train a model to forecast cloud cover based on a dataset from 2000-2010 designed for ML applications [7] using a classification threshold of 50% cloud cover (4 octas) to predict "cloudy" days.

Cloud cover is one of the many meteorological variables that are collected constantly by automated weather stations, as it plays a significant role in several aspects of atmospheric science. One of these roles cloud cover can play is being a forecasting variable, in which the overall cloud cover, usually represented by octas or eighths, as well as cloud height (low, medium, high level, or a mixture of these types) [8] can be used to predict variations in other atmospheric components, such as temperature and precipitation intensity. Climatological forecasts can also benefit from utilizing cloud cover by estimating the outgoing longwave radiation (OLR), which can be used to determine how moist or arid a region is, as a higher OLR indicates more radiation escaping to space, meaning there are less clouds in the region reflecting said radiation, and thus, less precipitation making it more arid [8]. Another role cloud cover plays is having a more direct effect on non-atmospheric aspects, such as increased cloud cover reducing temperature and increasing rainfall to promote crop growth [9] or even affecting the performance of

Major League Baseball players due to many factors, like how much the clouds block the Sun or if there is precipitation occurring during the game [10].

Cloud cover can be notoriously difficult to obtain widespread, accurate measurements for. Common forms of measuring include ground-based [11], and more recently, satellite [12], however both present problems. Measuring cloud cover from the ground is limited in scale and subject to error. Satellite measurements can be costly. The implementation of machine learning algorithms into cloud cover models then becomes crucial. Machine learning can act to reduce cost, increase accuracy, improve range from ground measurement, and bridge ground and satellite measurements to create a holistic cloud cover estimation [13]. To do this, we require both knowledge of atmospheric science, as well as machine learning, to both train algorithms and produce and validate understandable, verifiable data.

Literature Review

Due to the broad scope of cloud science in meteorology relating to cloud physics/microphysics, chemistry, geoengineering, classification, dynamics, and thermodynamics, this discussion will be concerned primarily with the area of study most relevant to this papers application: cloud/cloud cover and its effect on radiation/temperature. Several early studies found clouds to be a fundamental component in global radiative processes [14], [15], acting to absorb incoming shortwave and outgoing longwave radiation in the atmosphere. As a result, a wealth of subsequent papers have delved into the regional influence of clouds on radiative budgets and forcing in West Africa, the Indus River Valley, Eastern China, the Western Pacific, and the Arctic [16], [17], [18], [19], [20], to name a few. As computational capabilities and knowledge in boundary layer processes improved, doors opened up for the investigation of cloud cover on mesoscale radiative cooling [21], [22]: a significant influence on local low temperatures during calm, cloudless nights, with implications for shoulder season frosts/freezes and winter weather extrema. Beginning in the late 90s, the growing study of global warming led to retrospective analyses of cloud cover and its impact on climate changes, concluding that it’s closely correlated with climatic

changes in strength of high and low pressure centers [23], partially responsible for the variability in temperature changes across the US [24], [25], and in some parts of the world has been shown to be increasing [26]. This latter result supported the long leading theory that increased temperatures will result in increased cloud cover due to higher global evaporation rates, though some recent studies have contradicted this assumption, suggesting a positive climate feedback and reduced cloud cover due to increasing atmospheric stability [25], [27]. Additionally, with recently improved resolution of climate models, some climatic cloud projections have been developed [28]. Finally, the advent of machine learning techniques has ushered in a new tool for reanalysis of many prior ideas, ranging from radiative cooling [6], to cloud-specific forecast corrections [29] to, most recently, insights into the actual forecasting of clouds [13], [30].

With the divisive views on cloud cover effects under a changing climate and the novelty of deep learning capable of predicting cloud cover [30], there exists new opportunities in research to model previously complex processes with less computational power [31]. With ever increasing spatial resolution of cloud cover monitoring from remote sensing improvements, we must first develop a baseline cloud cover model from reanalysis data to act as a control for future predictive models operating on new, advanced data. The climate is now rapidly progressing into a new era, therefore we must first begin to understand the changes and effects by utilizing previous data availability to create a ground truth for cloud cover modeling.

By researching these relationships between cloud cover and radiation, including the utilization of different data analysis techniques for said research, many aspects of these relationships can be implemented in furthering our understanding of this topic, which includes both how it will aid and support others' research, and humankind in general. A recent study utilized a convolutional neural network (CNN) for post-processing cloud cover outputs from a numerical weather prediction model (ARPEGE), where some of their findings included that the CNN was proficient in identifying areas of extreme cloud cover (i.e., clear and overcast skies), as well as improving the accuracy of cloud cover forecasts over mountainous regions, but had issues with forecasting interposed cloud cover between clear

and overcast, and it overpredicts cloud cover in low-pressure systems [32]. From studies like these that use more recent data analysis tools like neural networks and machine learning, we can compare our own study involving these methods, which can lead to further developing the methods, and thus, improving future weather forecasts. In 2004, an article was published that investigated how observations of low cloud cover has changed over time due to newer methods to observe them, namely the implementation of Automated Surface Observing System (ASOS) stations, in which the authors found that the observed low cloud cover generally increased before the use of ASOS stations (1940s – 1980s), and then generally decreased afterward (1990s – 2002), where they theorized that a possible reason for this was long term multidecadal weather deviations [33]. Later in 2017, an article described in-depth a more recent index for low cloud cover, where the index implies that climatological changes in sea surface temperatures have a significant impact on low cloud cover [34], which goes back and supports the 2004 article's thoughts on long term atmospheric changes affecting low cloud cover.

Methodology

For this study, a cloud cover classification technique is used to estimate models' accuracy through a threshold of 50% cover, grouping 'cloudy' and 'not cloudy' environments. A number of established machine learning, ensemble, and deep learning models are tested to provide data on the performance of known techniques. These include support vector machines (SVM), k-nearest neighbors (KNN), XGBoost, adaboost, bagging, random forests, artificial neural networks (ANN), long short-term memory RNN (LSTM), convolutional neural networks (CNN), and bidirectional LSTM (BiLSTM). In addition to these, three novel methods developed within the last five years are applied, including extended, circular dilated CNN (CDIL) [35], and BiLSTM-CNN [36], and extended LSTM (xLSTM) [37] . A train-test split of 60/40, 70/30, and 80/20 are applied first before evaluating using classification metrics such as train accuracy, validation accuracy, test accuracy, f1 score, kappa, recall, precision, and area under the curve

(AUC). Python and scikit-learn tools were used in the coding application of these techniques. All data results can be seen in Figure 1.

Data

The data used is sourced from Kaggle and includes, among many other meteorological phenomena, daily cloud cover (oktas) in 18 European cities from 2000 to 2010 (3654 total timesteps). We use data from Basel in this study due to its centralized location as well as influence from complex surrounding terrain, which may provide more challenging forecasting conditions to train our models under. The dataset also provides a classification set of their own based on a combination of variables to determine appropriate "barbecue" weather, as it is designed to be used to teach and train machine learning models [7]. This portion of the dataset is not used here.

Machine Learning Techniques

SVM

Support vector machines are a supervised learning algorithm (requiring labelled data) that work to classify by defining a 'hyperplane' among a group of points. This plane is defined by a center line, the maximum margin, and two equidistant bounds, called positive and negative hyperplanes. The bounds exist as far apart as possible before making contact with any points in the dataspace so that area of the hyperplane is maximized. In linear SVM, this plane is straight through the points whereas nonlinear SVM allows them to divide irregular shaped data clusters. This nonlinear category can also be made linear by moving the data into a higher dimension using a kernel. SVM proved very capable in the majority of validation metrics, with the best results appearing in AUC with a score of ~0.96, the worst in Kappa with a score of ~0.75, and an average score across all metrics of ~0.89.

KNN

K-nearest neighbors is supervised and classified by taking individual points of data, considering k number of nearby points (this hyperparameter is specified beforehand, in this case, 7), and determining which group to be linked to based on the identity of the closest points. Higher values of k increase sensitivity to training data but run the risk of overfitting. In the case of the cloud cover data, the train and test accuracy were fairly comparable, at ~0.90 and ~0.88 respectively, suggesting that this value was appropriate. The highest metric was, as in SVM, AUC, with ~0.94 and the lowest was also Kappa at ~0.72. The average across all was 0.88, suggesting similarly adequate performance to SVM.

Ensemble Techniques

XGBoost

eXtreme Gradient Boosting (XGBoost), is an ensemble learning algorithm that utilizes multiple decision trees combined sequentially. The 'boosting' that occurs happens from one tree to the next; each tree is trained to correct errors and biases from the previous tree. While sequential decision trees would normally not be performant compared to other algorithms, XGBoost utilizes parallel processing to maintain quick processing speeds, especially on larger datasets. XGBoost is especially popular because it prevents overfitting through regularization and pruning. Overall, XGBoost performed well across all metrics on this dataset. A 100% accuracy on the training data signified a potential for overfitting, but XGBoost also received a 100% accuracy when used to predict on test data. This led XGBoost to be one of the highest scoring algorithms tested.

AdaBoost

Adaptive Boosting, or AdaBoost, is an ensemble technique that utilizes several simple decision trees, called decision stumps, to form a singular robust model. This is done by running through several boosting rounds, where if any of the weights output an error above 50%, they are reset and resampling occurs once

again, in which this occurs until each classifier has been properly weighted, where the weighted sum produces the final outcome prediction. The model had a train accuracy of 0.9211 and a test accuracy of 0.8933, and since the n_estimators of the model were set to 300, this could possibly be due to slight overfitting, or the model being better fitted for the training data over the test data. For the other evaluation metrics, all the values were more than 0.9, except for the kappa being 0.7715, giving the model an overall average of 0.9143.

Bagging

Bagging is a supervised ensemble classifier that uses sampling with replacement (bootstrap) to construct each classification member. These members are then each trained on the data and the plurality result for each timestep determines the model's prediction at that time. Higher number of members decrease variance and increase overfitting risk. In this study, a 50 member bagging model was used, which resulted in a train accuracy of 0.9996 and a test accuracy of 0.89, suggesting a slight overfit and that a smaller number of ensembles might have been ideal. Other results were quite good and kappa was the least performative at 0.75. The average was 0.90, making it the second best performing method tested after XGBoost.

Random Forest

Random forests are an ensemble model that are made of several decision trees that utilizes bootstrap sampling for training each decision tree, which is done until all trees are unpruned, and then the average of the ensemble outputs the prediction. Only certain features are selected and randomization occurs in order to decorrelate the decision trees, which aids in improving accuracy and reducing overfitting. The model had a train accuracy of 0.9740 and a test accuracy of 0.8728, and since the n_estimators of the model was set to 300, this could possibly be due to either slight overfitting or the model being better fitted for the training data over the test data. For the other evaluation metrics, all the values were more than 0.9,

with the kappa being the lowest at 0.9285 and the AUC being the highest at 0.9978, giving the model an overall average of 0.9513, indicating it performed very well.

Deep Learning

ANN

Artificial neural networks (ANN) can act as supervised or unsupervised machine learning algorithms and adapt to the use case (this data is labelled so it is supervised here). They are more capable at capturing complex relationships than simpler machine learning techniques. These models consist of several nodes that are directly linked together by edges, where a node receives a signal from an input edge, computes, and then sends the signal to an output edge, similar to how our brains send signals through neurons and synapses. The model had a train accuracy of 0.9772 and a test accuracy of 0.8906, in which this could possibly be due to either slight overfitting or the model being better fitted for the training data over the test data; however, this disparity could be reduced with further adjustments to the parameters. For the other evaluation metrics, all the values were more than 0.9, with the kappa being the lowest at 0.9345 and the AUC being the highest at 0.9934, giving the model an overall average of 0.9513, indicating it performed very well.

LSTM

Long Short-Term Memory (LSTM) is a form of Recurrent Neural Network (RNN) with the addition of memory, thus being able to retain information over multiple steps for longer timescale dependencies. This acts to solve a long-standing issue plaguing RNN called the vanishing problem. This problem is such that information is lost slowly, or vanishes, over an increasing amount of time steps from when the information is reduced. This limits traditional RNN's in situations where data is critical in making an accurate prediction from many timesteps prior. LSTM tackles this problem by introducing selective gates that can control a long-term memory cache when it is considered relevant to the current process. Three

gates are implemented: Input Gate, Forget Gate, and the Output Gate. The input gate controls what information is added to the LSTM's memory, the Forget Gate controls what is forgotten in the LSTM, and the Output Gate determines what information is output from the LSTM's memory to a current process. LSTM's are typically tailored towards NLP due to the natural formatting of words and memory requirements over the course of a text, however, they can be tailored to become a binary classifier for use on the weather dataset. Mostly, this entails modifying the input to be three dimensional and contain sequences of time series data, and formatting the output into a binary format. The LSTM had 0.9293 accuracy on the training data set after a termination of 10 epochs, but only a 0.5841 accuracy on the test data. This suggests potential overfitting of the model, which could be expected due to the relatively small dataset size, and its organization into time sequences. With binary classification, the bottom bar for accuracy is 0.5, which can be achieved without skill in prediction. While the LSTM is slightly better than random chance, it is significantly less performant than the traditional ML developed.

CNN

Convolution Neural Networks function like many networks do, with an added convolution layer. A convolution layer filters the input to extract out key information and make it machine-readable. This is primarily used in image detection and computer vision. Convolution layers can extract edges, shapes, and textures from an image and decode this in a way that a neural network will be able to recognize patterns from the image. The then decoded input, or image, is then passed to a fully connected neural network layer which can classify it based on the output from the convolutional layer(s). While this technique is not usually implemented in binary time-series classification, it achieved slightly better than random chance accuracy on the test data, at 0.6279. Not only was this better than random chance, it also performed slightly better than the LSTM. The CNN technique was designed by the addition of a convolution layer to the LSTM developed above, thus showing that the convolution layer can improve accuracy, even in non-image based datasets, likely from input-feature extraction helping to make patterns more accessible for the fully connected LSTM to classify into a binary prediction.

BiLSTM

Bidirectional Long Short-Term Memory networks are another variation of LSTM's that allow multidirectional information flow. Traditional LSTM's only allow sequential flow in one direction, without accounting for information that might occur after the sequence. BiLSTM's allow flow in either direction, allowing for enhanced context. This functions as two LSTM's, one run forward sequentially and the other ran backwards. This is then combined into a unified prediction, based on each individual LSTM prediction. BiLSTM had a similar, albeit slightly worse performance than a traditional LSTM. The test accuracy was 0.539, while the train accuracy was 0.9111, suggesting that BiLSTM suffers from the same overfitting than the traditional LSTM did, due to the small dataset and sequential ordering.

New Methods

xLSTM

Extended long short-term memory (xLSTM) are another variation of LSTM that utilizes exponential gating, modified memory structures (including matrix memory), and stabilization methods to improve over the base LSTM model, and even making it capable for language modeling. For the parameters utilized in this xLSTM model, the sequence length was set to 14 time steps (14 days) was set as a 2-week time period is ample enough to capture cloud evolution in a region and reduce overfitting by not making the time period too long, setting the target variable the same as the other methods of Basel's cloud cover, where the train-test split was set to 80/20, the number of LSTM layers was set to 2 with 64 hidden units (compromising between capacity and risk of overfitting), a cross-entropy loss function, an Adam optimizer (computes adaptive learning rates), a learning rate of 0.001, a batch size of 64, and a training duration of 20 epochs. While the parameterization of the xLSTM model is relatively scientifically sound, the model ended up performing quite poorly, where the validation output gave a train accuracy of 0.3146, a validation accuracy of 0.2683, and a test accuracy of 0.2541. An interesting observation about the validation metrics is that the AUC has the highest value of 0.6170, and when compared to the

accuracies, is much higher than those values, which could possibly be attributed to the model being able to distinguish the separate classes better than being accurate at predicting the cloud cover oktas with the relatively simple dataset provided. The average of the validation metrics of the xLSTM was about 0.2635, making it the worst performing model out of all the machine learning methods, possibly due to the simple nature of cloud cover in oktas, where the xLSTM performs best with more complicated data.

CDIL

Circular dilated CNN essentially derives from dilated CNNs, which apply a kernel to each convolutional layer with a gap parameter, effectively identifying longer range dependencies between pixels within the layer (in a graphical application). In this case, a sequential application, it sees each layer as a series of values across time steps. The circular nature of CDIL takes this convolutional layer and attaches the ends of the rows, allowing the kernel to act in a circular path and identifying relationships across greater timesteps by allowing for direct inclusion of the beginning and ends of the timestep sequence within one convolution output. Greater dilation coefficients (d, which determine how large the gap between grid cells in the kernel is) increase the relevance of this feature. The parameters used in this study are dilation constants 1, 2, and 3 with window sizes of 5, 10, and 20 time steps. These parameters combined in different permutations and then concatenated to derive validation results. The train accuracy is one in this case due to the complexity of the model, which would suggest some overfitting given a test accuracy of 0.76, but not drastic. The worst metric was kappa at 0.36, implying that CDIL is not ideal in this case at identifying the underlying pattern or predicting much better than random chance. Given that the overall average across all metrics was only 0.77, one might assume that the data was somewhat imbalanced. Some of the variance between kappa and average score in other tests would support this.

CNN-BiLSTM

This new method implemented a convolutional neural network layer and instead of a traditional fully connected LSTM layer, utilized a Bidirectional LSTM layer. First, the dataset was organized into a

3-dimensional dataset with sequences of individual time-series of length 10. Each sequence had 8 features over 10 timesteps. With an 80/20 train/test split implemented, this yielded a training dataset with the dimensions: (2914, 10, 8), and a test dataset of (731, 10, 8). Due to the binary classification, each corresponding target sample only had 1 feature for a training dimension of (2914, 1) and a test dimension of (731, 1). The model built was sequential with a convolutional layer, followed by a Bidirectional LSTM, and lastly a Dense layer for binary classification. The convolutional layer had 32 filters with a kernel size of 3. It utilized a Rectified Linear Unit activation function. This layer then passed into the Bidirectional layer, built with two LSTM's, each with 30 units and a tanh activation layer. Lastly, the output from the Bidirectional layer was input to a Dense layer with a unit of 1 and a sigmoid activation function. This was used for binary classification and effectively smushes the output from the LSTM into either a 1 or a 0 for output. The CNN-BiLSTM achieved a test accuracy of 0.6142, which is slightly better than random chance and the BiLSTM by itself. Overall, this is not a strong performance and remains similar in skill to the individual CNN, LSTM, and BiLSTM.

Results

Model	Train_accuracy	Val_accuracy	Test_accuracy	F1_score	Kappa	Recall	Precision	AUC
SVM	0.8842	0.8777	0.8925	0.9225	0.7474	0.9213	0.9237	0.9567
Random Forest	0.974	0.8878	0.8728	0.9829	0.9285	0.9687	0.9976	0.9978
AdaBoost	0.9211	0.9097	0.8933	0.9493	0.7715	0.9563	0.9424	0.9704
KNN	0.9014	0.8777	0.878	0.9115	0.715	0.9055	0.9176	0.9445
XGBoost	1	1	1	1	1	1	1	1
Bagging	0.9996	0.8577	0.8925	0.9227	0.7466	0.9239	0.9215	0.9466
ANN	0.9772	0.8947	0.8906	0.9853	0.9345	0.9894	0.9813	0.9934
LSTM	0.9293	0.5841	0.5841	0.3863	0	0.3029	0.2255	0.4826
Bi-LSTM	0.9111	0.539	0.539	0.3863	0	0.36	0.2188	0.4848
1D-CNN	0.8974	0.6279	0.6279	0.3863	0	0.2114	0.2164	0.4476
New Methods:								
xLSTM	0.3146	0.2683	0.2541	0.1874	0.0627	0.254	0.1502	0.617
CDIL	1	0.7124	0.7593	0.8407	0.362	0.9171	0.776	0.7591
CNN-BiLSTM	0.9118	0.6142	0.6142	0.3863	0	0.2743	0.2365	0.4775

Figure 1. Validation Results Table

Overall across the dataset, simpler algorithms performed the best across all metrics used. Average across all metrics of 0.89 and models was while more complex algorithms tended to overfit and lack skill in prediction. The best performances were attained by SVM, KNN, XGBoost, and Bagging with high performance across all metrics. Deep learning networks generally failed at skilled prediction with this dataset with frequent overfitting and no test accuracy higher than 0.61 (achieved by CNN-BiLSTM) nor an overall score greater than 0.44. The average across all deep-learning models was 0.45. An exception to this is the new technique developed, a Circular Dilated Convolution (CDIL) neural network, which achieved skill in prediction across all metrics rivaling the highest scores (>0.76 test accuracy with 0.77 overall).

Ensemble algorithms were a mixed bag on this dataset with two having the best performance out of all algorithms; XGBoost and Bagging, and the other two, AdaBoost and RandomForest, performing poorly, with accuracy in the 0.6 range.

Performance Evaluation Measures

Many evaluation metrics were used to test model skill including: accuracy, precision, recall, F1 score, Cohen's kappa, and the area under the ROC curve (AUC). Accuracy is widely implemented across model performance due to its intuitive nature as it’s simply the proportion of correctly predicted outcomes out of the total number. However, this can present problems in biased datasets as a model could have high accuracy while maintaining low skill in predicting a select few critical outcomes. Cohen’s Kappa is similar to accuracy but crucially deliminates how often a correct prediction could occur by random chance, thus a better metric on unbalanced or biased datasets. Precision measures the amount of positive predictions the model makes out of the total number of positive outcomes, while recall measures the amount of total positive outcomes the model correctly predicts. The F1 score is then the harmonic mean of precision and recall and is useful to balance a dataset with making an equal number of false positive and false negative predictions. Lastly, the AUC measures how well a model can correctly identify a

positive outcome over a negative outcome. This is critical for performance on a binary classification dataset such as this, when the model needs to correctly identify a positive weather day over negative weather days.

Discussion

Models that performed well in test accuracy tended to perform well in all other metrics, yielding overall scores comparable to that value. Exceptions to this include 1D-CNN and CNN-BiLSTM, both of which had test accuracy values higher than the average across validation tests. This may have resulted in their inadequacy in dealing with imbalanced data. Consistently, the lowest performing metric across all models was the Cohen's kappa, likely due to ambiguous signals in weather. Recall and precision, and thusly F1 score held up well with the strong performing models. Overall the metrics support the conclusion that simple ML models performed the strongest, with the highest scores across all metrics in SVM, and KNN. Additionally, models derived from simple models such as ensemble and bagging algorithms had a wide spread, with the two best performing models overall, XGBoost and Bagging, but also some poorly performing ensemble models with RandomForest and AdaBoost. Deep learning was largely lacking skill in prediction with this dataset, with only CDIL as an exception, but continuing to lack abilities in kappa.

Conclusion

This study aimed to utilize several machine learning techniques on forecasting cloud cover, in which these methods were evaluated with the threshold of 50% (binary of either no clouds or clouds) from the 2000-2010 Basel, Switzerland weather dataset, along with several verification metrics being calculated for each model for validation, as well as to compare and contrast the methods. Overall, the simpler machine learning algorithms tended to perform better across all the validation metrics than the more complex techniques, as those tended to have poor prediction skill and succumb to overfitting. Of all

the techniques, XGBoost performed the best with a 100% in the accuracies, as well as all the validation metrics, making it a good fit for modeling cloud cover, and more specifically whether or not it will be cloudy, which could possibly be attributed to XGBoost performing well with this specific dataset, or that it actually overfitted the data even though the test accuracy was also 100%. While the machine learning methods, ensemble methods, and deep learning methods all generally performed well, there was some variation in the performance of the ensemble methods and especially the deep learning methods, whereas the machine learning methods were more consistent with their metrics. The more complex models, especially the new methods, performed worse than the simpler methods, most likely due to overfitting the data, as well as the dataset lacking complexity. Overall, this study highlights the utility of simple machine learning techniques in fittingly simple datasets, and in this case binary cloud cover prediction. Future work that could be done includes inputting complex cloud cover datasets, such as satellite observations, into the more complex models to better understand and improve these methods.

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